

African Folktales SLM Dataset

Data Card

Dataset Version 1.0

1. Dataset Summary

African Folktales SLM Benchmark

The African Folktales SLM Dataset was developed to support the training and evaluation of a domain-specific Small Language Model (SLM) focused on African folktale generation, cultural preservation, and storytelling.

The dataset is intended to help explore how small language models can learn narrative structures, themes, cultural regions, moral lessons, and storytelling patterns associated with African folktales.

The dataset currently contains 24 folktale documents, 38 training prompt-story pairs, and 10 test prompts. The dataset metadata identifies the corpus as synthetic and provides a CC-BY-4.0 license.

2. Motivation

Traditional African storytelling has historically relied heavily on oral transmission. As oral storytelling declines and many narratives remain poorly documented or inaccessible in digital formats, there is a risk of losing important cultural knowledge, values, moral lessons, and narrative traditions.

This dataset supports our broader goal of developing an AI-powered system that can preserve, organize, study, and generate African folktale-style narratives.

The dataset was created as an initial benchmark for experimenting with domain-specific language modeling and should be viewed as a foundation that can be expanded with larger, community-verified collections.

3. Intended Use

The dataset is intended for:

- Training small language models for folktale generation.
- Studying African storytelling structures.
- Natural Language Processing research.
- Educational applications.
- Cultural heritage technology.
- Creative storytelling applications.

- Evaluating domain-specific language models.
- Experimenting with retrieval, fine-tuning, LoRA, and other model adaptation techniques.

The project brief describes the broader objective as building an SLM capable of generating traditional stories, representing diverse regional mythologies, and maintaining oral storytelling structures.

4. Out-of-Scope Uses

The dataset should not be used to:

- Claim that the included stories represent all African cultures.
- Treat synthetic stories as historically verified oral traditions.
- Make generalized claims about African communities.
- Present AI-generated stories as authentic traditional accounts without verification.
- Infer cultural beliefs or practices from the dataset without additional evidence.
- Replace cultural historians, community elders, researchers, or indigenous knowledge holders.
- Establish historical facts solely from generated model outputs.

5. Dataset Composition

The dataset consists of the following components:

6. Dataset Distribution

Themes

Theme	Documents
Trickster	4
Origin Myth	4
Moral Tale	4
Hero Journey	4
Animal Fable	4
Community Wisdom	4
Total	24

This balanced distribution reduces the risk of the initial corpus being dominated by a single narrative category.

Cultural Regions

Region	Documents
West Africa	6
East Africa	6
Southern Africa	5

Central Africa	5
Diaspora	2
Total	24

These distributions were obtained from the corpus exploration performed on documents.csv.

7. Data Schema

documents.csv

Field	Type	Description
document_id	String	Unique identifier for each document
title	String	Title of the folktale
text	String	Main folktale text
theme	String	Narrative category
culture_region	String	Broad cultural/geographical region
origin	String	Stated origin information
source_url	String	Source/reference URL
license	String	Licensing information

train_prompts.csv

Field	Type	Description
PromptId	Integer	Unique prompt identifier
prompt	String	User instruction for generating a story
theme	String	Expected narrative theme
culture_region	String	Target cultural region
document_id	String	Related source document
reference_story	String	Reference story used for training /evaluation

test_prompts.csv

Field	Type	Description
PromptId	Integer	Unique prompt identifier
prompt	String	Story-generation instruction
theme	String	Target narrative theme
culture_region	String	Target cultural region

The test prompts intentionally do not expose their reference stories.

8. Data Generation and Provenance

The current benchmark is explicitly described in the metadata as a synthetic folktale corpus. Therefore, the stories should not automatically be interpreted as direct transcriptions of traditional oral narratives.

Each document includes provenance-related fields such as origin, source_url, and license, allowing future versions of the dataset to improve traceability.

The dataset uses a CC-BY-4.0 license according to the supplied metadata.

Future dataset versions should prioritize the inclusion of:

- Recorded oral histories.
- Community-contributed stories.
- Published African folklore with appropriate permissions.
- Anthropological and cultural research sources.
- Indigenous-language material.
- Community-verified translations.
- Clear attribution and provenance information.

The project specification recommends sourcing culturally rich material from recorded oral histories, translated anthropological texts, and published African literature while accounting for cultural nuance and dialect representation.

9. Data Processing

The dataset is processed for use in language-model training and retrieval experiments.

The current pipeline includes:

1. Loading the supplied CSV files.
2. Inspecting dataset structure and metadata.
3. Checking themes and cultural regions.
4. Normalizing textual content for retrieval.
5. Creating word-level TF-IDF representations.
6. Creating character-level TF-IDF representations.
7. Combining textual similarity with theme and regional metadata.
8. Retrieving relevant source documents for prompts.
9. Creating candidate story outputs.
10. Evaluating retrieval performance against known training examples.

The retrieval experiment uses both word and character-level TF-IDF representations and metadata-aware scoring.

10. Quality Checks

The dataset was evaluated for whether the retrieval system could correctly identify the source document associated with each known training prompt.

Document retrieval accuracy: 100% (38/38 training prompts).

However, this result should not be interpreted as 100% model accuracy. It only demonstrates that the retrieval pipeline successfully identified the associated document for the available training examples.

The mean Levenshtein distance between the complete source documents and their corresponding reference stories was 112.42 in the local analysis.

The actual Kaggle evaluation uses a hidden reference set, meaning the local retrieval result does not directly represent the final competition score.

11. Evaluation

The benchmark evaluates generated stories using mean character-level Levenshtein distance, where a lower score indicates greater similarity to the hidden reference stories.

The evaluation setup encourages the model to produce stories that are semantically and structurally aligned with the expected reference narratives.

Because the hidden reference stories are unavailable, local evaluation is primarily used to:

- Test document retrieval.
- Validate preprocessing.
- Compare candidate-generation strategies.
- Detect data leakage or processing errors.
- Compare baseline and improved models.

12. Bias and Representation

Africa represents thousands of cultures, languages, ethnic communities, and storytelling traditions. A data set containing only 24 documents cannot adequately represent this diversity.

Potential sources of bias include:

Geographic bias

West and East African stories are currently slightly more represented than Southern and Central African stories, while the diaspora category has only two documents.

Linguistic bias

The current benchmark is primarily text-based and does not adequately represent the large number of indigenous African languages.

Cultural bias

Broad regional labels may simplify complex cultural identities and should not be interpreted as precise representations of individual communities.

Synthetic-data bias

Because the corpus is synthetic, some narratives may reflect the assumptions or stylistic choices of the dataset creators rather than authentic community storytelling traditions.

Translation bias

Future multilingual versions may introduce additional meaning loss when stories are translated between indigenous languages and English.

13. Ethical Considerations

Cultural preservation requires more than simply collecting stories and training a model.

The project therefore follows five guiding principles:

Cultural preservation:

Use AI to help preserve and increase access to African storytelling traditions.

Inclusivity:

Expand representation across regions, cultures, communities, and languages.

Fairness:

Avoid allowing a small number of regions or cultural groups to dominate the dataset.

Transparency:

Clearly communicate whether a story is sourced, adapted, synthetic, or AI-generated.

Responsible AI:

Avoid presenting generated narratives as historically authentic without supporting evidence.

14. Limitations

The current dataset has several important limitations:

11. Small dataset size: Only 24 documents and 38 training examples are currently available.
12. Synthetic corpus: The dataset does not yet provide comprehensive evidence of authentic oral traditions.
13. Limited linguistic diversity: Indigenous African languages are not adequately represented.
14. Limited geographical coverage: The dataset cannot represent the full diversity of African cultures.

15. Potential cultural simplification: Broad regional labels may hide important differences between communities.
16. Limited evaluation data: Only 10 test prompts are available in the supplied benchmark.
17. Potential overfitting: The small training set creates a significant risk of overfitting when fine-tuning a language model.

Because the training dataset is only 38 prompt/reference pairs, our experiments indicate that full fine-tuning would likely overfit; therefore, parameter-efficient methods such as LoRA/QLoRA are more appropriate candidates for experimentation.

15. Recommended Future Improvements

Future versions of the dataset should expand both quantity and cultural authenticity.

Priority improvements include:

- Increase the number of folktales substantially.
- Collect stories from additional African countries.
- Include indigenous African languages.
- Work with cultural experts and community knowledge holders.
- Add community verification.
- Record the original language of each story.
- Preserve multiple translations where available.
- Add detailed provenance metadata.
- Distinguish traditional, adapted, translated, and synthetic stories.
- Add human evaluation for cultural authenticity.
- Add safety and cultural-misrepresentation checks.
- Create balanced representation across regions and narrative traditions.

16. Responsible Use Statement

This dataset should be considered an experimental benchmark and foundation for cultural-preservation research, not a comprehensive digital representation of African folklore.

Model outputs generated from this dataset should be clearly labelled as AI-generated or AI-assisted unless the underlying narrative has been independently verified as an authentic traditional source.

Researchers and developers using the dataset should preserve source attribution, respect applicable licensing requirements, avoid cultural stereotypes, and seek appropriate community involvement when expanding

g the dataset.

17. Dataset Version

Item	Value
Current version	1.0
Dataset type	Synthetic African folktale corpus
Documents	24
Training prompt/story pairs	38
Test prompts	10
Themes	6
Cultural-region categories	5
License	CC-BY-4.0
Primary task	African folktale generation
Evaluation metric	Mean character-level Levenshtein distance

18. Conclusion

The African Folktales SLM Dataset provides a small but structured starting point for investigating domain-specific language models for cultural storytelling and preservation.

While the current corpus is too small and synthetic to represent the diversity of African folklore, its structured themes, regional metadata, source-document relationships, and prompt-story pairs provide a useful foundation for experimentation.

The long-term objective is to evolve the benchmark into a larger, multilingual, culturally informed dataset developed with stronger provenance, expert review, and community participation.

This approach will allow the resulting SLM to move beyond simply generating stories that “sound African” toward producing narratives that are more culturally grounded, transparent, respectful, and useful for education, research, and cultural preservation.